Electric Forces: Guided Notes · ANSWER KEY

Chemistry · Unit 1B · Level 3 (closer to the class work) · goes with L1.7-L1.9 (Coulomb's Law, Neutral Interactions, Static Interaction Review)

Name: _____

Use the class slides and your reading. Key words:

electric force | *attract* | *repel* | *Coulomb's law* | *distance* | *amount of charge* | *scientific notation* | *exponent* | *polarized* | *neutral*

A. Attract or repel (L1.7, L1.9)

1. Opposite charges **attract**. Like charges **repel**.

2. Two balloons are both rubbed on hair. Predict what happens when they hang near each other, and explain.
They repel because both gained electrons and are negative (like charges).

B. Coulomb's law: proportional reasoning (L1.7)

3. More charge makes the force **stronger**. More distance makes the force **weaker**.

4. **Charge rule: multiply the force by the same number as the charge.**

> Example: The force is 6 mN. One charge doubles (× 2). New force = 6 × 2 = 12 mN.

a) Force is 5 mN. One charge triples (× 3). New force = **15 mN**

b) Force is 8 mN. One charge is cut in half (÷ 2). New force = **4 mN**

5. **Distance rule: square the change, then divide.**

> Example: The force is 36 mN. The distance doubles (× 2). 2 × 2 = 4, so new force = 36 ÷ 4 = 9 mN.
> If the distance gets smaller, multiply instead: half the distance → force × 4.

a) Force is 20 mN. Distance doubles. New force = **5 mN**

b) Force is 18 mN. Distance triples. New force = **2 mN**

c) Force is 3 mN. Distance is cut in half. New force = **12 mN**

C. Scientific notation (L1.7)

6. **Write the number out: move the decimal to the right as many places as the exponent.**

> Example: $2 \times 10^3 = 2{,}000$ $4.5 \times 10^4 = 45{,}000$

a) 6×10^5 = **600,000**

b) 2.3×10^3 = **2,300**

c) Which is more charge: 3×10^{12} e or 8×10^9 e? **3×10^{12} e (bigger exponent)**

D. Neutral objects (L1.8)

7. A neutral object near a charged object becomes **polarized** .

8. Draw a negative balloon next to a neutral piece of paper. Show where the paper's electrons move. Draw an arrow for the force.

Balloon with − signs; paper with − on the far side and + on the near side; arrow pulling the paper toward the balloon.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*